Auto Loans & Borrowing — Cheat Sheet

Financial Algebra · Unit 5: Auto Loans & Borrowing

Essential vocabulary

Installment loan
A loan repaid in equal scheduled payments over a fixed term

Loan term
The length of time to repay a loan

Down payment
Cash paid up front that reduces the amount borrowed

Amortization
Paying off a loan through regular payments that cover interest first and then principal

Amortization schedule
A table showing each payment's interest, principal, and remaining balance

Finance charge
The total dollar cost of credit, including interest and certain fees

Total of payments
The sum of all scheduled payments over the life of a loan

Upside down (negative equity)
Owing more on a loan than the item is worth

Depreciation
The decline in an asset's value over time

Lease
A contract to use a vehicle for a set term and mileage in exchange for payments, without owning it

Residual value
The estimated value of a leased vehicle at the end of the lease

Money factor
The lease version of an interest rate; multiply by 2,400 to estimate an equivalent APR

Collateral
An asset pledged to secure a loan that the lender can take if payments stop

Cosigner

Key formulas

Monthly rate i	$\frac{\text{APR}}{12}$
Payment	$\text{PMT} = \frac{P \cdot i}{1 - (1 + i)^{(-n)}}$
Max loan for a payment	$P = \frac{\text{PMT}(1 - (1 + i)^{(-n)})}{i}$
Total of payments	PMT × n
Finance charge	total of payments − amount financed
Lease payment	$\frac{\text{cap} - \text{residual}}{\text{months}}$ + (cap + residual) × MF
APR from money factor	MF × 2,400

Amortization row

Interest	balance × i
Principal	payment − interest
New balance	balance − principal

Worked example

1. $22,000 at 6.9%
2. 48 months: $525.81/month → total $25,238.88
3. 84 months: $330.83/month → total $27,789.72
4. Longer term costs about $2,551 more

A person who agrees to repay a loan if the borrower does not

Prepayment
Paying more than the required payment to reduce the principal faster

Total cost of ownership
All costs of having a vehicle over time: price or payments, interest, insurance, fuel, maintenance, fees, and lost value

Common misconceptions

- Lowest payment = cheapest loan.
- Interest is the same every month.
- Add-ons cost only their sticker price.
- Loan approval = affordability.

Questions to ask before deciding

- What is the out-the-door price?
- What APR do I qualify for elsewhere?
- What is the total of payments?
- How long until I owe less than the car is worth?
- What will insurance, fuel, and maintenance cost?

Decision tips

- Get preapproved before visiting the dealer.
- Negotiate price, then APR, then term.
- Avoid financing add-ons.
- Keep the term as short as your budget allows.

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